

MATH 55 FINAL EXAM, PROF. SRIVASTAVA
DECEMBER 19, 2024, 11:40AM–2:30PM, 150 WHEELER.

1. Select true or false for each of the following by filling in a circle. There is no need to provide an explanation.

(a) (3 points) The compound proposition

$$(\forall x((p(x) \rightarrow q(x)) \wedge (\neg p(x) \rightarrow q(x))) \rightarrow \forall x q(x))$$

is a tautology.

☐ True

☐ False

- (b) (3 points) If A, B, C are nonempty sets, $f : A \rightarrow B$ is one-to-one, and $g : B \rightarrow C$ is one-to-one, then $g \circ f : A \rightarrow C$ must be one-to-one.

☐ True

☐ False

- (c) (3 points) For any three integers a, b, c , if $\gcd(a, b) = 1$ and $\gcd(b, c) = 1$ then $\gcd(a, c) = 1$.

☐ True

☐ False

- (d) (3 points) If a, b are nonzero integers then $(a + b)^5 \equiv a^5 + b^5 \pmod{5}$.

☐ True

☐ False

- (e) (3 points) If X and Y are nonnegative random variables and $\mathbb{P}(X \leq Y) = 1$ then $V(X) \leq V(Y)$, where $V(\cdot)$ denotes the variance of a random variable.

☐ True

☐ False

[Scratch space below]

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2. Assume a and b are irrational numbers with $a < b$.

(a) (4 points) Prove that the function $f : (0, 1) \rightarrow (a, b)$ defined by $f(x) = a + (b - a)x$ is a bijection.

(b) (6 points) Prove that there exists an irrational number c such that $a < c < b$.

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3. (a) (2 points) Prove that if x is an integer then $x^2 \equiv 0 \pmod{3}$ or $x^2 \equiv 1 \pmod{3}$.

(b) (3 points) Prove that if x and y are integers such that

$$x^2 + y^2 \equiv 3xy \pmod{3}$$

then $x \equiv y \equiv 0 \pmod{3}$. You may use the statement of part (a).

(c) (5 points) Prove that the equation $x^2 + y^2 = 3xy$ has no *positive* integer solutions.
(hint: proof by contradiction. Use the well-ordering principle and (b)).

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4. For $n \geq 1$, let a_n denote the number of subsets of $\{1, \dots, n\}$ which do *not* contain two consecutive numbers (i.e., do not contain i and $i + 1$ for some i).

(a) (2 points) What are a_1 and a_2 ?

(b) (6 points) Derive a recurrence relation for a_n , $n \geq 3$.

(c) (2 points) Use your recurrence relation to compute the value of a_{10} .

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5. (a) (5 points) Prove that if G is a connected simple graph on n vertices with $n+1$ edges, then G must contain at least two distinct (i.e., not equal) cycles as subgraphs.

- (b) (5 points) Prove that if G is any simple graph on n vertices with $n+1$ edges, then it must contain at least two distinct cycles. You may use the statement of part (a) even if you did not prove it.

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6. Let n be an even positive integer.

(a) (6 points) Prove that if G is a simple graph on n vertices in which every vertex has degree at least $n/2$, then G must be connected.

(b) (4 points) Give an example of a disconnected simple graph G on n vertices in which every vertex has degree exactly $(n/2) - 1$.

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7. Suppose Nikhil draws a hand of 5 cards uniformly at random from a deck of 52 (13 cards $A, 2, \dots, 10, J, Q, K$ of each of the four suits: hearts, spades, diamonds, clubs).

(a) (3 points) What is the probability that the hand is a flush, that is, five cards of the same suit?

(b) (3 points) What is the probability that the hand contains at least one ace?

(c) (3 points) What is the probability that the hand is a flush *and* contains at least one ace?

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8. Consider the following experiment: Oski rolls a fair six-sided die and observes a uniformly random outcome $D \in \{1, 2, \dots, 6\}$. After that, Oski flips D independent fair (i.e., equally likely to be heads or tails) coins. Let C be the number of heads observed.

(a) (3 points) What is $\mathbb{P}(C = 5|D = 6)$?

(b) (4 points) What is $\mathbb{P}(D = 6|C = 5)$?

(c) (3 points) For what value of x is $\mathbb{P}(D = x|C = 5)$ maximized?

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9. Assume that there are 500 lockers labeled $\{1, \dots, 500\}$ at the Recreational Sports Facility (RSF) at Cal (which I highly recommend, if you haven't been there). Assume each one has a combination lock whose code is an independent uniformly random 4-digit string with digits in $\{0, 1, \dots, 9\}$. Call an unordered pair of lockers *vulnerable* if their locks have the same code.

(a) (3 points) For $i, j \in \{1, \dots, 500\}$ with $i < j$, let E_{ij} denote the event that the pair $\{i, j\}$ is vulnerable. Calculate $\mathbb{P}(E_{12})$.

(b) (2 points) Are the events E_{12} and E_{34} independent? Justify your answer.

(c) (2 points) Are the events E_{12} and E_{23} independent? Justify your answer.

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- (d) (4 points) Let X be a random variable equal to the number of vulnerable pairs. Calculate the expected value $\mathbb{E}X$.

- (e) (5 points) Calculate the variance $V(X)$.